ESP32 WROOM

Use boot control pins with caution: 0, 2, 5, 12, 15

IO0: Avoid device connections. Can be used a stat LED.
IO2: Avoid external pullups – will cause bootloader fail.
IO5: Has builtin pullup at POR.
IO12: Avoid external pullups – will cause bootloader fail.
IO15: Has builtin pullup at POR.

ADC2 is unavailable when WiFi is enabled.

VDD Range: 3.0V – 3.6V

USB-C

Open jumper to isolate shield

Reset and Boot

Boot Mode Configuration

Pin	Run	Bootloader	Default
IO0	1	0	1
IO2	x	0	0
IO5	1	x	1
IO12	0	0	0
IO15	x	x	1

IO12 = MDTI, IO15 = MDTO

If DTR is LOW, toggling RTS from HIGH to LOW resets to run mode.
If RTS is HIGH, toggling DTR from LOW to HIGH resets to bootloader.

LEDs

Open jumper to disable LED
Open jumper to disable LED

Switched Power

Vin Max: 6.0V
Iout Max: 700mA
Vdo: 120mV @ 330mA
Output Discharge: 530Ω Typ

microSD

MICRO-SD-PUSH-PUSH-EXT-PIN

System Power

Vin Max: 6.0V
Iout Max: 700mA
Vdo: 120mV @ 330mA
Output Discharge: 530Ω Typ

LiPo Battery Charger

Charge Current: 500mA
Open jumper to disable LED

Fuel Gauge

Open jumpers to disable pull-ups

Strapping Pin Isolation

74HC4066_QUAD_BILATERAL_SWITCH_QFN

IMU

CS: I²C/SPI mode selection
1: SPI idle mode / I²C communication enabled
0: SPI communication mode / I²C disabled

VDD Range: 1.71V – 3.6V
VDDIO Range: 1.62V – 3.6V

Magnetometer

SPI_CS:
Chip Select line for SPI (active low)
Tie to VDDIO for I²C Interface

VDD Range: 2.8V – 3.6V
VDDIO Range: 2.8V – 3.6V

Qwiic Bus Isolation

74HC4066_QUAD_BILATERAL_SWITCH_QFN

Breakouts and Connectors

QWIIC_RIGHT_ANGLE
Vertical Qwiic Connector

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Special Instructions

TITLE: ${PROJECTNAME}

Design by: ${DESIGNER}

REV: ${REV}

Date: ${CURRENT_DATE}

Sheet: ${#}/${##}

Sheet: /
File: wroom.kicad_sch

Size: User | Date: | Rev:
KiCad E.D.A. eeschema 7.0.6 | Id: 1/1